MACHINE WHITE PAPERS

Fadal VMC 4020-906-1 — Pistol Whiskers

CNC vertical mill for precision milling, drilling, and contouring on prototypes through medium production runs. 20" W × 48" L × 24" H workpiece envelope.



Pistol Whiskers · CNC vertical machining center

The Fadal 4020 is the workhorse of American medium-envelope CNC milling — rigid, well-understood, and forgiving with the harder alloys. Pistol Whiskers is our go-to for prototype work, repair parts, and small-to-medium production runs where the part fits inside the envelope. It handles complex pocketing, milled features, and multi-face setups without flinching on the alloys that would chatter marginal equipment.

What Pistol Whiskers is for

Pistol Whiskers is the machine that makes a job economical before anyone has decided the job is worth doing. A 20" × 48" × 24" envelope with a CAT-40 spindle and a 21-tool changer is not exotic, and that is exactly the point: it is fast to set up, cheap to run, and forgiving enough that a one-off repair part or a three-piece prototype doesn't have to justify tying up a larger machine. Most parts that come through this shop as "can you just make one of these" start here.

Pick this machine when

- The part fits inside 20" × 48" × 24" and is prismatic — brackets, plates, manifold blocks, housings, adapters.
- You need one, or five, and the setup cost matters more than the cycle time.
- It's a repair or reverse-engineered part and the geometry is still being confirmed against a worn sample.
- The job needs multi-face work with a tool list that fits comfortably inside 21 pockets.

When it's the wrong machine

Anything round that wants to be turned — that belongs on a lathe, and we have seven. Parts over 20" in width go to Loco Motive, which has more than twice the envelope and a heavier column to match. High-volume small parts are usually wrong here too: the ATC and the CAT-40 spindle are general-purpose, and a job running hundreds of pieces will cost you more in cycle time than it saves in setup.

From the floor

The CNC-88 control on the original Fadal builds is older than a lot of the prints we get, and that's a feature — it is transparent, it is repairable, and nobody on the floor is waiting on a service contract to make a change at the machine. Where it costs us is in high-speed contouring on 3D surfaces, so complex surfacing gets programmed conservatively rather than pushed.

Pick your industry

Fadal VMC 4020 for Oil & Gas

Full spec sheet, typical oil & gas parts, alloys, tolerance expectations, documentation approach.

Fadal VMC 4020 for Aerospace

Full spec sheet, typical aerospace parts, alloys, tolerance expectations, documentation approach.

Fadal VMC 4020 for Military & Defense

Full spec sheet, typical defense parts, alloys, tolerance expectations, documentation approach.

Fadal VMC 4020 for Industrial & General Manufacturing

Full spec sheet, typical industrial parts, alloys, tolerance expectations, documentation approach.

Frequently asked

What does B&R Productions use the Fadal VMC 4020-906-1 for?

When is the Pistol Whiskers the right machine for my part?

When is the Pistol Whiskers the wrong machine?

What industries does B&R machine on the Fadal VMC 4020?

How do I get a quote for work on the Fadal VMC 4020?

Have a project?